

**Comsats University Islamabad,
Attock Campus**



Human Computer Interaction

Submitted to: Dr. Tahira Sadaf

Submitted by: Zain Habib

Registration no: FA22-BSE-020



Submission Date

December 12th, 2025

User Evaluations

Scenario 1: First-time Purchase Flow

- "You want to buy 'Frozenda Brazile' coffee. Find it and add one bag to your cart."
- "Go to your cart and proceed to checkout."
- "Complete the checkout process using online payment."

Scenario 2: Cart Management

- "You have three items in your cart. Change the quantity of one item to ."
- "Remove one item from your cart."
- "See what your total would be before checking out."
- "Does same product which is added to cart and some navigation and you checked that item in browsing area has visible that it has been added to cart"

Scenario 3: Order Confirmation & Tracking

- "You just placed an order. Find out if it was successful."
- "Find how to track your order after purchase."
- "Check how long delivery will take."

Scenario 4: Ratings System

- "You just press the rating button that it change the color"
- "after REFRESH the page see that rate is still there"

Expert Evaluations

I am doing the expert evaluation on the base of Heuristic Rules which is proposed by Herbert A. Simon which is often described as a "Right Thumb Rule". On the bases of this right thumb rule here is the Expert Evaluation:

Aesthetic and Minimalist Design (Heuristic #8) & Consistency and Standards (Heuristic #4)

The One Coffee Product.png screen suffers from severe visual clutter and information duplication.

The customer review "It was on amazing Coffee Website..." by "Larry wapsi" is repeated verbatim four times. This destroys credibility and provides zero useful information.

The page layout attempts to present product info, CTAs, and reviews in a single, dense column without clear visual hierarchy.

Effects:

This fundamentally undermines user trust and makes critical product information (like the review) meaningless. It also violates the principle that every piece of information must carry its weight.

Match Between System and the Real World (Heuristic #2)

Multiple instances of nonsensical or incorrect data that breaks user trust and comprehension.

The order summary has a critical arithmetic error. Product Price: 49\$, Quantity: , Tax: 8\$ should equal $(49 \times 8) + 8 = 547\$$. The displayed Total Amount: 49 \$ is wildly incorrect.

The pricing error is catastrophic for conversion. Users will not complete a purchase with obviously wrong financial information.

Visibility of System Status (Heuristic # 1)

The system provides incorrect or misleading status information.

The message "Special Order has Been Delivered Successfully" is shown immediately after payment confirmation. The correct status should be "Order Confirmed" or "Order Placed." "Delivered" indicates the final step in the fulfillment journey, which hasn't happened.

The header icon reads "Cart (0)", but the screen clearly displays three items in the cart. The counter does not reflect the actual system state.

Hypothesis based on Dataset

Hypothesis # 1

Store Location and Transaction Quantity

Sr#	store_location	transaction_qty
1	Lower Manhattan	2
2	Lower Manhattan	2
3	Lower Manhattan	2
4	Lower Manhattan	1
5	Lower Manhattan	2
6	Lower Manhattan	1
7	Lower Manhattan	1
8	Lower Manhattan	2
9	Lower Manhattan	1
10	Lower Manhattan	2

Hypothesis (one-tailed)

“Customers at Lower Manhattan (population) buy **more items per transaction** (measurement) than customers at Hell’s Kitchen (feature).”

Null Hypothesis (H_0)

“There is no difference in transaction quantities between Lower Manhattan and Hell’s Kitchen.”

IV: store_location (Lower Manhattan vs Hell’s Kitchen)

DV: transaction_qty (numeric; items per transaction)

Experimental Design

Take a sample of transactions (e.g., 20 from each store). Group by store_location and compare transaction_qty using an independent-samples t-test.

Condition A (Lower Manhattan): transaction_qty values for Lower Manhattan transactions.

Condition B (Hell’s Kitchen): transaction_qty values for Hell’s Kitchen transactions.

Hypothesis # 2

Product Category and Unit Price

Sr#	product_category	unit_price
1	Coffee	3.00
2	Tea	3.10
3	Drinking Chocolate	4.50
4	Coffee	2.00
5	Tea	3.10
6	Bakery	3.00
7	Coffee	2.00
8	Coffee	2.00
9	Coffee	4.25
10	Drinking Chocolate	3.50

Hypothesis (one-tailed)

“Items in the **Coffee** category (population) have a higher unit price (measurement) than items in the **Tea** category (feature).”

Null Hypothesis (H₀)

“There is no difference in unit price between Coffee and Tea items.”

IV: product_category (Coffee vs Tea)

DV: unit_price (numeric)

Experimental Design

Select equal samples of Coffee and Tea transactions (e.g., 20 each). Compare unit_price between the two groups with an independent t-test.

Condition A (Coffee): unit_price values for product_category = Coffee.

Condition B (Tea): unit_price values for product_category = Tea.

Hypothesis # 3

Time of Day and Transaction Quantity

Sr#	transaction_time	transaction_qty
1	07:06:11	2
2	07:08:56	2
3	07:14:04	2
4	07:20:24	1

5	07:22:41	2
6	07:22:41	1
7	07:25:49	1
8	07:33:34	2
9	07:39:13	1
10	07:39:34	2

Hypothesis (one-tailed)

“Early-morning customers (population) (transactions before 08:00) purchase **fewer items per transaction** (measurement) than late-morning customers (feature) (transactions at or after 08:00).”

Null Hypothesis (H₀)

“There is no difference in transaction quantities between early-morning and late-morning transactions.”

IV: time_period (Early morning < 08:00 vs Late morning ≥ 08:00) — derived from transaction_time

DV: transaction_qty (numeric)

Experimental Design

Split transactions into two groups by transaction_time: before 08:00 and 08:00 or

later. Take a sample (e.g., 20 per group) and compare transaction_qty with an independent t-test.

Condition A (Early morning): transactions with transaction_time < 08:00.

Condition B (Late morning): transactions with transaction_time ≥ 08:00.

Hypothesis # 4

Product Type and Purchase Frequency (Proportion)

Sr#	product_type
1	Ethiopia Rg (Coffee)
2	Spicy Eye Opener Chai Lg (Tea)
3	Dark chocolate Lg (Drinking Chocolate)
4	Our Old Time Diner Blend Sm (Coffee)
5	Spicy Eye Opener Chai Lg (Tea)
6	Oatmeal Scone (Bakery)
7	Our Old Time Diner Blend Sm (Coffee)
8	Columbian Medium Roast Sm (Coffee)
9	Latte Rg (Barista Espresso)
10	Dark chocolate Rg (Drinking Chocolate)

Hypothesis (one-tailed)

“Barista Espresso items (population) are purchased **more frequently** (measurement) than Drinking Chocolate items.”

Null Hypothesis (H_0)

“There is no difference in purchase frequency between Barista Espresso and Drinking Chocolate.”

IV: product_type (Barista Espresso vs Drinking Chocolate)

DV: purchase frequency (count / proportion of transactions)

Experimental Design

From the dataset count how many transactions include Barista Espresso and how many include Drinking Chocolate (use same sampling window). Compare proportions (e.g., test whether proportion of Barista Espresso purchases > proportion of Drinking Chocolate purchases) using a chi-square / proportion z-test.

Condition A (Barista Espresso): transactions where product_type = Barista Espresso.

Condition B (Drinking Chocolate): transactions where product_type = Drinking Chocolate.

Hypothesis based on Activities

Hypothesis 1

Order Confirmation Clarity

Users who see "Order has been delivered" in the confirmation screen (current design) will have significantly higher confusion rates about order status and be more likely to contact customer service than users who see "Order confirmed" (redesigned version).

Null Hypothesis: There will be no significant difference in confusion rates or customer service contact likelihood between users exposed to "delivered" versus "confirmed" terminology.

Independent Variable: Confirmation screen terminology ("Delivered" vs "Confirmed")

Dependent Variables: % of users who correctly identify order status

Hypothesis 2

Product Review Credibility

Users viewing duplicate identical reviews (current design) will report significantly lower trust in product ratings and be less likely to purchase compared to users viewing varied, unique reviews (redesigned version).

Null Hypothesis: There will be no significant difference in perceived trustworthiness or purchase intent between duplicate and varied review presentations.

Independent Variable: Review presentation format

Dependent Variable: --Product trust rating --Purchase intent score

Hypothesis 3

Cart Management Effectiveness

Users attempting to modify item quantities using the "Edit" button (current design, requiring page navigation) will have higher frustration levels, more interaction steps, and higher cart abandonment rates than users using inline +/- controls (redesigned version).

Null Hypothesis: There will be no significant difference in frustration, interaction steps, or cart abandonment between the two cart editing methods.

Independent Variable: Cart quantity modification method

Dependent Variables: Number of clicks/steps to change quantity